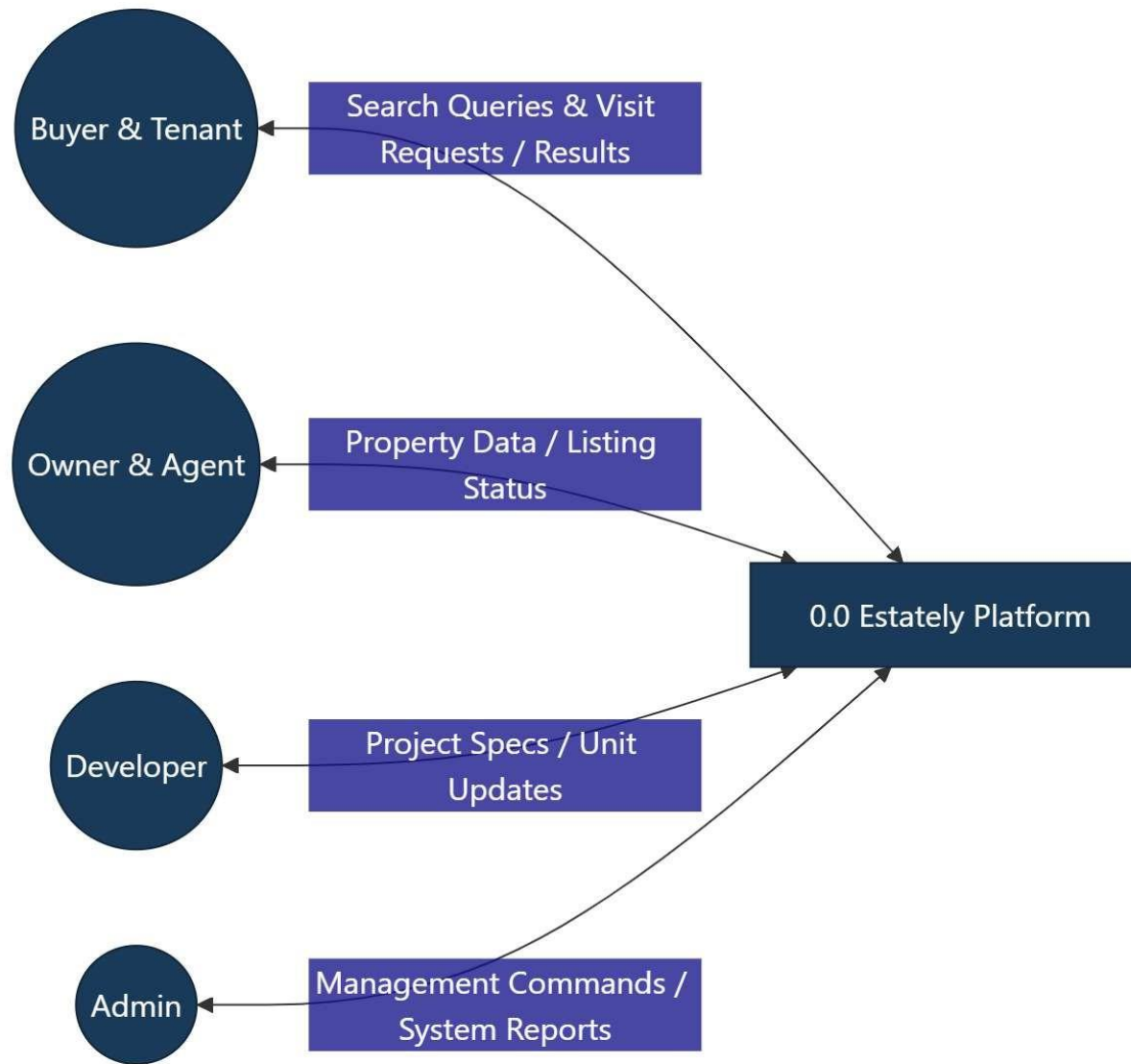


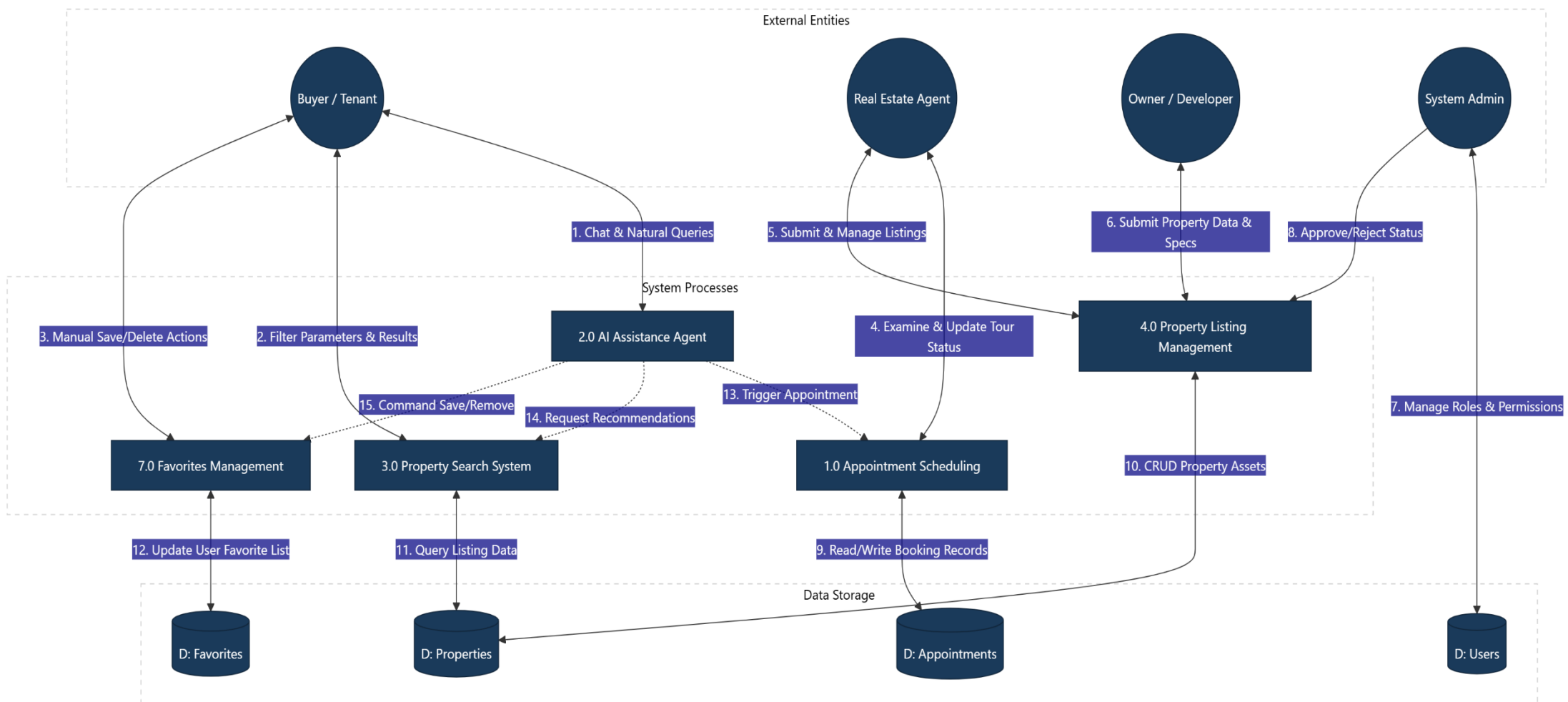
1. DFD Level 0: Global Context Diagram

This diagram provides a high-level overview of the Estately system boundary, showing the primary interactions between external entities (Buyers, Owners, Developers, and Admins) and the platform as a single process. It illustrates the flow of search queries, property data, and management commands into the system, as well as the resulting search results and status notifications.



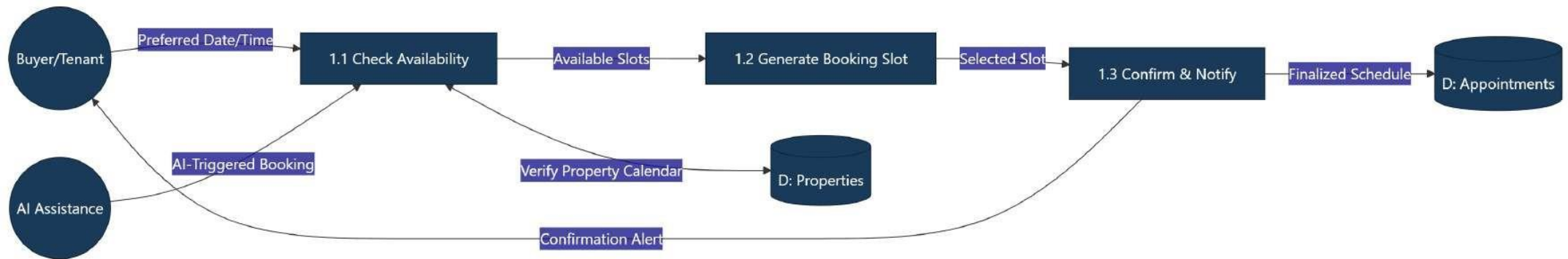
DFD Level 1: Functional Decomposition and Agentic Interaction

This level breaks the platform into its core modular processes, including the **AI Assistance Agent**, **Property Search System**, and **Favorites Management**. It highlights the agentic workflow where the AI orchestrates tasks like triggering visit requests and recommending properties. It also shows the centralized **Favorites Management** process handling both manual and AI-driven save requests before interacting with the database.



DFD Level 2: Appointment Scheduling System

This diagram details the step-by-step process of booking a property visit. It shows the interaction between the User (or AI Agent) and the scheduling module to check property availability, generate a valid booking slot, and finalize the schedule in the **Appointments** database.



5. DFD Level 2: Property Listing and Status Management

This diagram illustrates the lifecycle of a property listing from creation to approval. It captures how owner or agent data is validated and then moderated by the **Admin**, who has the authority to approve, reject, or mark a listing as "Sold" before the final record is committed to the **Properties** database.



3. DFD Level 2: AI Assistance Agentic Workflow

This detailed view explores the internal logic of the AI agent. It demonstrates how the agent analyzes user intent and context to delegate specific actions: generating personalized recommendations through the search system, orchestrating appointments, or commanding the favorites management process to up

